

HYPERBOLIC GEOMETRY — EXERCISES FROM LECTURE 3

Exercise 3.A

Show that the stereographic projection centred at $(0, 0, -1)$ on the plane $\{z = 0\}$ induces a Riemannian isometry $\mathcal{H}^2 \rightarrow \mathbb{D}^2$.

Exercise 3.B

Let $p, q \in \mathcal{H}^2$.

- (i) Show that there is $g \in \mathrm{O}^+(2, 1)$ such that $gp = q$ and $gq = p$.

Hint: Consider $(p - q)^\perp$.

- (ii) Show that any isometry of $T_p\mathcal{H}^2$ is induced by an element of $\mathrm{O}^+(2, 1)$.

- (iii) Using (i) and (ii), deduce that $\mathrm{O}^+(2, 1) = \mathrm{Isom}(\mathcal{H}^2)$.

Property (i) is transitivity of the action $\mathrm{O}^+(2, 1) \curvearrowright \mathcal{H}^2$, and (ii) is its isotropy.

Exercise 3.C

Fix a k -vector space V .

- (i) Given two distinct points $p \neq q \in \mathbb{P}(V)$, prove that there is a unique projective line through p and q .
- (ii) Show that two distinct projective lines in $\mathbb{P}(V)$ intersect in a unique point.
- (iii) Let $\omega \in \mathbb{P}(V)$, and let $\mathbb{P}(L)$ be a projective line in $\mathbb{P}(V)$ not containing ω . The projection on $\mathbb{P}(L)$ based at ω is the map $\pi : \mathbb{P}(V) \setminus \{\omega\} \rightarrow \mathbb{P}(L)$ defined as follows: given $p \in \mathbb{P}(V) \setminus \{\omega\}$, let (p, ω) be the unique projective line through p and ω , and let $\pi(p)$ be the unique point of $(p, \omega) \cap \mathbb{P}(L)$. Given a projective line $\mathbb{P}(L')$ not containing ω , show that the restriction $\pi|_{\mathbb{P}(L')} : \mathbb{P}(L') \rightarrow \mathbb{P}(L)$ is a projective isomorphism (i.e. is the projective map induced by a linear isomorphism). Deduce that π preserves the cross-ratio.

Exercise 3.D

Let a, b, c, d, e be five distinct points in $\mathbb{P}^1(k)$.

- (i) Express the following cross-ratios in terms of $[a, b, c, d]$:

$$(\alpha) \quad [a, c, b, d] \qquad (\beta) \quad [d, b, c, a] \qquad (\gamma) \quad [b, a, c, d].$$

- (ii) Prove the equality $[a, b, d, e] = [a, b, c, e] \cdot [a, c, d, e]$.

- (iii) Assume that $k = \mathbb{R}$. Suppose that a, a', b, c, d', d occur in this order in $\mathbb{P}^1(\mathbb{R})$. Show that $[a', b, c, d'] \geq [a, b, c, d]$.

Exercise 3.E

Given a, b, c, d four distinct elements of $\mathbb{P}^1(\mathbb{C}) = \mathbb{C} \cup \{\infty\}$, prove that

$$[a, b, c, d] = \frac{c - a}{b - a} \cdot \frac{b - d}{c - d}.$$

Exercise 3.F

Let z_1, z_2, z_3, z_4 be four distinct points in $\mathbb{P}^1(\mathbb{C}) = \mathbb{C} \cup \{\infty\}$.

- (i) Prove that the circle or line $\mathcal{C}(z_1, z_2, z_4)$ through z_1, z_2, z_4 is orthogonal to the circle or line $\mathcal{C}(z_1, z_3, z_4)$ through z_1, z_3, z_4 if and only if

$$\Re[z_1, z_2, z_3, z_4] = 0.$$

N.B. If z_1, z_2, z_4 are colinear, or if one of them is ∞ , then $\mathcal{C}(z_1, z_2, z_4)$ is a line; in all other cases, it is a circle.

- (ii) Generalise (i) to characterise the property that $\mathcal{C}(z_1, z_2, z_4)$ and $\mathcal{C}(z_1, z_3, z_4)$ meet at an angle ϑ .

Exercise 3.G

- (i) Show that the projection $\mathbb{R}^3 \setminus \{0\} \rightarrow \mathbb{P}^2(\mathbb{R})$ restricts to an isometry (in the sense of metric spaces) $\mathcal{H}^2 \rightarrow \mathbb{K}^2$.
- (ii) Determine the geodesics of $\mathcal{H}^2$.